

Project Design Phase Solution Architecture

Date	02-07-2026
Team ID	23A51A4422
Project Name	Laptop Request Process in ServiceNow
Maximum Marks	4 Marks

Solution Architecture:

The Laptop Request Process is built on the ServiceNow platform using a layered architecture approach. Employees interact with the system through a Service Catalog item that captures request details such as Laptop Type and Business Justification. The application logic layer processes the request, validates the submitted information, initiates the manager approval workflow using Flow Designer, and automatically generates a Catalog Task for the IT Hardware team after approval. The platform also sends automated email notifications to inform the requester about the completion of the laptop provisioning process.

• User Interaction Layer

The Service Catalog provides a self-service interface where employees can request a laptop by selecting the required laptop type, entering a business justification, and submitting the request through an intuitive online form.

• Business Logic Layer

Flow Designer, Catalog Variables, and Approval Actions work together to validate the submitted request, route it to the employee's manager for approval, generate the required fulfillment task, and automate the complete request workflow.

• Data & Service Layer

ServiceNow tables such as `sc_request`, `sc_req_item`, and `sc_task` store request, requested item, and catalog task information. The platform also uses Email Notifications, Assignment Groups, and Workflow Execution Logs to support communication, task management, and process monitoring throughout the request lifecycle.

Solution Architecture Diagram:

